localCloudManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"localCloudManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `localCloudManagement` service of the LocalCloudManager system.

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `localCloudManagement` service interface of the LocalCloudManager system, which provides the operator console API of the local cloud: overview, registry and data views, orchestration rules, consumers, inter-cloud state, sysMonitor history and subscriptions, plant descriptions, logs and container lifecycle.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `localCloudManagement` service interface

This document provides the Interface Design Description IDD to the *localCloudManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `localCloudManagement` service interface in detail. Both in terms of its operations (Section 3) and its information model (Section 4).

2.1 Interface registration

LocalCloudManager listens on two connectors. The Arrowhead connector (port 8491, HTTPS with mutual TLS) is registered in the ServiceRegistry with the interface template `generic_https` and policy `CERT_AUTH`, base path `/api`. The console connector (port 8490, plain HTTP, published on the host's loopback address only) serves the browser console with the same routes plus the console-only ones; it is not registered. The `operations` property lists the operations below with their method and path relative to the base path (path values are appended to the registered path):

Operation	Method	Path
`overview`	GET	`/overview`
`serviceRegistry Management`	GET	`/service-registry-management`
`dataManager`	GET	`/datamanager`
`consumers`	GET	`/consumers`

Operation	Method	Path
interCloud	GET	/intercloud
sysMonitorHistory	GET	/sysmonitor
sysMonitorSubscriptions	GET	/sysmonitor/subscriptions
subscribeSysMonitor	PUT	/sysmonitor/subscriptions
unsubscribeSysMonitor	DELETE	/sysmonitor/subscriptions
plantDescriptions	GET	/plantdescriptions
createPlantDescription	POST	/plantdescriptions
replacePlantDescription	PUT	/plantdescriptions
updatePlantDescription	PATCH	/plantdescriptions
removePlantDescription	DELETE	/plantdescriptions
orchestrationRules	GET	/orchestration/rules
createOrchestrationRule	POST	/orchestration/rules
removeOrchestrationRule	DELETE	/orchestration/rules
logs	GET	/logs
lifecycle	POST	/containers

generalManagement is registered separately (base path /localcloudmanager/general/mgmt, operation get-config, GET /get-config?keys=..., Sysop only).

2.2 Access

Arrowhead connector (8491). A request needs the caller's client certificate and Authorization: Bearer IDENTITY-TOKEN//<token> of the same system (common outsourced authentication filter; 401 otherwise, with the common error body {"errorMessage", "errorCode", "exceptionType", "origin"}). A registered operation is allowed for Sysop, the systems of lcm.api.whitelist and systems with a Consumer-Authorization grant of the operation (403 otherwise); other /api paths only for Sysop and the whitelist. The console-only paths (/api/session, /health, the UI) are 404 there.

```
GET /api/overview HTTP/1.1
Host: cloud-monitor-api:8491
Authorization: Bearer IDENTITY-TOKEN//<identity token>
```

Listing 1: An invocation on the Arrowhead connector.

Console connector (8490). Host and Origin must be allowed console names (403 "Host or Origin is not an allowed console address"; Origin null is refused); with a configured proxy secret, /api requests must carry X-Console-Proxy (403 "Not forwarded by the console"); without a configured password every /api request is 503. Every /api route except /api/session needs the session cookie ah_console (401 "Console login required"). Other paths serve the static console (index.html for unknown paths). Responses carry X-Frame-Options: DENY, Content-Security-Policy: frame-ancestors 'none', X-Content-Type-Options: nosniff and Referrer-Policy: no-referrer.

3 Service Operations

The interfaces of the `localCloudManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `localCloudManagement` SD document, as well as input and output types. The examples use the console connector; on the Arrowhead connector the same request carries the identity token instead of the cookie.

For `POST`, `PUT`, `PATCH` and `DELETE` the header `X-Arrowhead-Action: confirm` is required (428 `"Missing action confirmation header"`). Request bodies are `application/json` and at most 64 KiB (larger: 500 `"Request body is too large"`; malformed: 500 `"Invalid JSON body"`); an empty body counts as `{}`. Responses are `application/json` with `Cache-Control: no-store`. Errors of the routes are answered with `{"error": "<text>"}` and the status codes `400 Bad Request` (invalid input), `401 Unauthorized`, `403 Forbidden`, `404 Not Found` (`"Not found"` for an unknown route), `405 Method Not Allowed`, `409 Conflict` (stored entry changed since it was read), `415 Unsupported Media Type` (login without JSON), `428 Precondition Required`, `429 Too Many Requests` (login locked), `500 Internal Server Error`, `502 Bad Gateway` (a provider request failed) and `503 Service Unavailable` (no authorized provider found by orchestration, or no console password). An error of a provider is passed on with the provider's status and error message.

3.1 GET /api/session

Operation: session
Output: Session

Whether the request carries a valid console session (no session needed). Console connector only.

A successful call returns `200 OK` with the output in the body.

```
GET /api/session HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 2: An session invocation.

```
{
  "authenticated": true,
  "configured": true
}
```

Listing 3: An session response.

3.2 POST /api/session

Operation: login
Input: Login

Logs in with the console password. The body must be `application/json` (415 otherwise). On success the response sets `ah_console=<token>; HttpOnly; SameSite=Strict; Path=/; Max-Age=28800`. A wrong password is 401 `{"error": "Wrong password"}`; a locked client is 429 with `Retry-After` and `{"error": "Too many failed logins; try again later"}`. Other methods on this path are 405. Console connector only.

A successful call returns `204 No Content`.

```
POST /api/session HTTP/1.1
Host: 127.0.0.1:8490
Content-Type: application/json

```

```
{
  "password": "<console password>"
}
```

Listing 4: An login invocation.

3.3 DELETE /api/session

Operation: logout
Output: Session

Ends the console session; the response clears the cookie (`Max-Age=0`). Console connector only.

A successful call returns `200 OK` with the output in the body.

```
DELETE /api/session HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 5: An logout invocation.

```
{
  "authenticated": false
}
```

Listing 6: An logout response.

3.4 GET /health

Operation: health
Output: Health

Liveness and ServiceRegistry registration state; no session needed. Console connector only (404 on the Arrowhead connector).

A successful call returns `200 OK` with the output in the body.

```
GET /health HTTP/1.1
Host: 127.0.0.1:8490
```

Listing 7: An health invocation.

```
{
  "status": "ok",
  "system": "LocalCloudManager",
  "registration": {
    "registered": true,
    "instanceId": "LocalCloudManager|generalManagement|1.0.0",
    "instanceIds": ["LocalCloudManager|generalManagement|1.0.0",
      "LocalCloudManager|localCloudManagement|1.0.0"],
    "error": null,
    "checkedAt": "2026-11-09T10:00:00.000Z"
  }
}
```

Listing 8: An health response.

3.5 GET /api/features

Operation: features
Output: Features

Which optional console views are enabled (not registered).

A successful call returns `200 OK` with the output in the body.

```
GET /api/features HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 9: An features invocation.

```
{
  "localCloudConfiguration": true,
  "chipletFloorplans": false,
  "managerName": "LocalCloudManager"
}
```

Listing 10: An features response.

3.6 GET /api/overview

Operation: overview
Output: View

Overview: containers of the managed Compose project (Docker proxy), ServiceRegistry inventory with runtime state, registry management, naming check, CRL validity, brokers and sysMonitor flows, the last 60 samples and the registration state. A failing source is reported in `errors` and the corresponding `cloud` flag.

A successful call returns `200 OK` with the output in the body.

```
GET /api/overview HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 11: An overview invocation.

```
{
  "collectedAt": "2026-11-09T10:00:00.000Z",
  "cloud": {"name": "arrowhead-v52-test-cloud", "release": "5.2.2",
            "registryReachable": true, "dockerReachable": true},
  "summary": {"at": "...", "running": 21, "containers": 23,
              "registeredSystems": 19, "registeredServices": 64,
              "unhealthy": 0, "incidents": 0, "unregisteredServices": []},
  "devices": [ ... ], "systems": [ ... ], "services": [ ... ],
  "registryManagement": { ... },
  "naming": {"valid": true, "violations": []},
  "revocation": {"thisUpdate": "...", "nextUpdate": "...",
                 "hoursRemaining": 70, "state": "ok"},
  "infrastructure": [ ... ], "orchestrations": [ ... ],
  "containers": [{"id": "0123456789ab", "name": "...", "service": "sysmonitor",
                  "displayName": "SysMonitor", "image": "...",
                  "state": "running", "status": "Up 2 hours (healthy)",
                  "health": "healthy", "created": "...", "managed": true}],
  "history": [ ... ],
  "registration": { ... },
  "errors": {"docker": null, "arrowhead": null, "naming": null}
}
```

Listing 12: An overview response.

3.7 GET /api/service-registry-management

Operation: serviceRegistryManagement
Output: View

Registry inventory grouped by device: systems with their services and runtime state, runtime-only containers and a summary.

A successful call returns `200 OK` with the output in the body.

```
GET /api/service-registry-management HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 13: An serviceRegistryManagement invocation.

```
{
  "collectedAt": "...",
  "cloud": { ... },
  "devices": [{"name": "Device1", "addresses": ["10.0.0.1"], "metadata": {},
               "registered": true, "systems": [ ... ],
               "deployed": true, "serviceCount": 5}],
  "runtimeOnly": [ ... ],
  "summary": {"registeredDevices": 1, "deployedDevices": 1,
              "registeredSystems": 19, "deployedSystems": 18,
              "registeredServices": 64, "deployedServices": 60,
              "runtimeOnly": 0},
  "errors": { ... }
}
```

Listing 14: An serviceRegistryManagement response.

3.8 GET /api/datamanager?limit=*{limit}*

Operation: dataManager
Output: View

DataManager `dataHistorian` records of all systems and services (latest `limit` entries, 1 to 1000, default 240), the latest entry per measurement and numeric series.

Parameters: *limit*.

A successful call returns `200 OK` with the output in the body.

```
GET /api/datamanager?limit=240 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 15: An dataManager invocation.

```
{
  "collectedAt": "...",
  "count": 1,
  "latest": [{"id": "RandomDataProducer|randomNumberProducer|urn:...:value|1000",
              "name": "urn:arrowhead:rdp:value", "time": 1000, "unit": null,
              "value": {"v": 42}, "record": { ... },
              "system": "RandomDataProducer", "service": "randomNumberProducer"}],
  "entries": [ ... ],
  "series": [{"name": "urn:arrowhead:rdp:value",
              "points": [{"time": 1000, "value": 42, "unit": null}],
              "system": "RandomDataProducer", "service": "randomNumberProducer",
              "sourceCloud": null,
              "label": "RandomDataProducer . value - local"}]
```

```
}
```

Listing 16: An dataManager response.

3.9 GET /api/consumers

Operation: consumers
Output: View

Per provider service, the authorized consumers (ConsumerAuthorization policies) and the orchestrated consumers (DSO history, GateTunnel sessions, SimpleStore rules, DSO plant rules).

A successful call returns `200 OK` with the output in the body.

```
GET /api/consumers HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 17: An consumers invocation.

```
{
  "collectedAt": "...",
  "localCloud": "TestCloud",
  "errors": [],
  "rows": [{"provider": "ServiceRegistry", "serviceDefinition": "temperature",
            "providerCloud": "TestCloud",
            "authorized": [{"consumerCloud": "TestCloud", "level": "MGMT",
                            "consumers": "Consumer", "scoped": []}],
            "orchestrated": [{"consumer": "Consumer", "consumerCloud": "TestCloud",
                              "via": "SimpleStore rule (priority 1)"}]}]
}
```

Listing 18: An consumers response.

3.10 GET /api/intercloud

Operation: interCloud
Output: View

Neighbour clouds, relay links of Gatekeeper and GateTunnel, GateTunnel sessions and the inter-cloud publishing of RandomDataProducer.

A successful call returns `200 OK` with the output in the body.

```
GET /api/intercloud HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 19: An interCloud invocation.

```
{
  "collectedAt": "...",
  "localCloud": "TestCloud",
  "neighbourClouds": ["GoCloud|Company"],
  "neighbourError": null,
  "relay": {"gatekeeper": {"connected": true}, "gatetunnel": {"connected": true}},
  "sessions": {"sessions": []},
  "producer": {"status": "publishing"}
}
```

Listing 20: An interCloud response.

3.11 GET /api/sysmonitor?system={system}&limit={limit}

Operation: sysMonitorHistory
Output: View

Systems known to SysMonitor (`sysMonitorHistory`) with their latest observations, and the observations of one system (`system`, default the first; `limit` 1 to 1000, default 120).

Parameters: *system*, *limit*.

A successful call returns `200 OK` with the output in the body.

```
GET /api/sysmonitor?system=DataManager&limit=120 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 21: An sysMonitorHistory invocation.

```
{
  "collectedAt": "...",
  "retentionDays": 7,
  "storedObservations": 3,
  "systems": [{"systemName": "DataManager", "observationCount": 3,
               "latest": {"echo": {"status": "alive"}}}],
  "selectedSystem": "DataManager",
  "observations": [ ... ]
}
```

Listing 22: An sysMonitorHistory response.

3.12 GET /api/sysmonitor/subscriptions

Operation: sysMonitorSubscriptions
Output: View

SysMonitor providers, subscriptions and AMQP intake state (`sysMonitorManagement`).

A successful call returns `200 OK` with the output in the body.

```
GET /api/sysmonitor/subscriptions HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 23: An sysMonitorSubscriptions invocation.

```
{
  "collectedAt": "...",
  "defaultSubscription": ["echo"],
  "providers": [{"systemName": "DataManager"}],
  "subscriptions": [],
  "intake": {"connected": true}
}
```

Listing 24: An sysMonitorSubscriptions response.

3.13 PUT /api/sysmonitor/subscriptions/*{system}*

Operation: subscribeSysMonitor
Input: SysMonitorSubscriptionRequest
Output: View

Sets the SysMonitor subscription of a provider system (`[A-Za-z0-9_-]+`); `enabled` defaults to true. The response is SysMonitor's.

Parameters: *system*.

A successful call returns `200 OK` with the output in the body.

```
PUT /api/sysmonitor/subscriptions/DataManager HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "observationTypes": ["echo", "status"],
  "enabled": true
}
```

Listing 25: An subscribeSysMonitor invocation.

```
{
  "systemName": "DataManager",
  "enabled": true
}
```

Listing 26: An subscribeSysMonitor response.

3.14 DELETE /api/sysmonitor/subscriptions/*{system}*

Operation: unsubscribeSysMonitor
Output: Removal

Removes the SysMonitor subscription of a provider system.

Parameters: *system*.

A successful call returns `200 OK` with the output in the body.

```
DELETE /api/sysmonitor/subscriptions/DataManager HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
```

Listing 27: An unsubscribeSysMonitor invocation.

```
{
  "removed": "DataManager"
}
```

Listing 28: An unsubscribeSysMonitor response.

3.15 GET /api/plantdescriptions

Operation: plantDescriptions
Output: View

Plant descriptions (first 200) with the registered systems and services offered as choices; `registryError` when the inventory failed.

A successful call returns `200 OK` with the output in the body.

```
GET /api/plantdescriptions HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 29: An plantDescriptions invocation.

```
{
  "collectedAt": "...",
  "entries": [{"id": 1, "plantDescription": "test", "active": false, ... }],
  "count": 1,
  "systems": ["Consumer", "ServiceRegistry"],
  "services": [{"provider": "ServiceRegistry", "serviceDefinition": "temperature",
                "interfaces": ["generic_https"]}],
  "registryError": null
}
```

Listing 30: An plantDescriptions response.

3.16 POST /api/plantdescriptions

Operation: createPlantDescription
Input: PlantDescription
Output: PlantDescription

Creates a plant description in the PlantDescriptionEngine; the body is forwarded.

A successful call returns `201 Created` with the output in the body.

```
POST /api/plantdescriptions HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "plantDescription": "Line 1",
  "active": false,
  "include": [],
  "systems": [],
  "connections": []
}
```

Listing 31: An createPlantDescription invocation.

```
{
  "id": 2,
  "plantDescription": "Line 1",
  ...
}
```

Listing 32: An createPlantDescription response.

3.17 PUT /api/plantdescriptions/*{id}*

Operation: replacePlantDescription
Input: PlantDescription
Output: PlantDescription

Replaces a plant description (numeric identifier; 404 without one).
Parameters: *id*.
A successful call returns `200 OK`.

```
PUT /api/plantdescriptions/2 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{ ... }
```

Listing 33: An replacePlantDescription invocation.

3.18 PATCH /api/plantdescriptions/*{id}*

Operation: updatePlantDescription
Input: PlantDescription
Output: PlantDescription

Updates a plant description.
Parameters: *id*.
A successful call returns `200 OK`.

```
PATCH /api/plantdescriptions/2 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "active": true
}
```

Listing 34: An updatePlantDescription invocation.

3.19 DELETE /api/plantdescriptions/*{id}*

Operation: removePlantDescription
Output: Removal

Removes a plant description.
Parameters: *id*.
A successful call returns `200 OK` with the output in the body.

```
DELETE /api/plantdescriptions/2 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
```

Listing 35: An removePlantDescription invocation.

```
{
  "removed": 2
}
```

Listing 36: An removePlantDescription response.

3.20 GET /api/orchestration/policies

Operation: authorizationPolicies
Output: PolicyView

Local-cloud management policies of ConsumerAuthorization (editable) and provider-owned local policies (read-only); not registered.

A successful call returns `200 OK` with the output in the body.

```
GET /api/orchestration/policies HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 37: An authorizationPolicies invocation.

```
{
  "entries": [ ... ],
  "readOnlyEntries": [ ... ],
  "collectedAt": "..."
}
```

Listing 38: An authorizationPolicies response.

3.21 POST /api/orchestration/policies

Operation: grantAuthorizationPolicy
Input: PolicyChange
Output: ManagementPolicy

Grants a local management policy (201), or replaces the stored policy of the same provider and target when `expected` matches it (200). 400 for an invalid policy, 409 for an unregistered provider or service, a policy that exists without `expected`, a changed policy or duplicate targets, 502 when a replacement could not complete.

A successful call returns `201 Created or 200 OK` with the output in the body.

```
POST /api/orchestration/policies HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "policy": {"provider": "ServiceRegistry", "targetType": "SERVICE_DEF",
             "target": "temperature",
             "defaultPolicy": {"policyType": "WHITELIST",
                               "policyList": ["Consumer"]}},
  "expected": null
}
```

Listing 39: An grantAuthorizationPolicy invocation.

```
{
```

```
  "instanceId": "MGMT|LOCAL|ServiceRegistry|SERVICE_DEF|temperature",
  ...
}
```

Listing 40: An grantAuthorizationPolicy response.

3.22 DELETE /api/orchestration/policies/*{instanceId}*

Operation: revokeAuthorizationPolicy
Input: PolicyChange
Output: Removal

Revokes a local management policy (URL-encoded instance identifier) when `expected` matches the stored policy: 404 unknown, 428 without `expected`, 409 changed.

Parameters: *instanceId*.

A successful call returns `200 OK` with the output in the body.

```
DELETE /api/orchestration/policies/MGMT%7CLOCAL%7CServiceRegistry%7CSERVICE_DEF%7Ctemperature HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "expected": { ... }
}
```

Listing 41: An revokeAuthorizationPolicy invocation.

```
{
  "removed": true,
  "id": "MGMT|LOCAL|ServiceRegistry|SERVICE_DEF|temperature"
}
```

Listing 42: An revokeAuthorizationPolicy response.

3.23 GET /api/orchestration/rules

Operation: orchestrationRules
Output: View

SimpleStore rules of the registered consumers, the orchestration catalog (producers with their interoperable consumers) and the aggregated DynamicServiceOrchestration history.

A successful call returns `200 OK` with the output in the body.

```
GET /api/orchestration/rules HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 43: An orchestrationRules invocation.

```
{
  "catalog": {"producers": [{"instanceId": "ServiceRegistry|temperature|1.0.0",
                             "provider": "ServiceRegistry",
                             "serviceDefinition": "temperature",
                             "version": "1.0.0", "interfaces": [ ... ],
                             "compatibleConsumers": ["Consumer"]}],
```

```
                "consumers": [{"name": "Consumer", "version": null, "x509": true,
                               "interfaceTemplates": ["generic_https"]}]},
  "rules": [{"id": "123e4567-...", "consumer": "Consumer",
             "serviceInstanceId": "ServiceRegistry|temperature|1.0.0", "priority": 1}],
  "count": 1,
  "dynamic": [{"consumer": "Consumer", "serviceDefinition": "temperature",
               "count": 1, "kind": "local",
               "providers": [{"provider": "ServiceRegistry", "cloud": null}], ... }],
  "dynamicError": null,
  "collectedAt": "..."
}
```

Listing 44: An orchestrationRules response.

3.24 POST /api/orchestration/rules

Operation: createOrchestrationRule
Input: OrchestrationRule
Output: View

Creates a SimpleStore rule after checking that the producer is still offered and the consumer is interoperable with it (400 invalid, 409 stale or incompatible). The response is SimpleStore's.

A successful call returns `201 Created`.

```
POST /api/orchestration/rules HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "consumer": "Consumer",
  "serviceInstanceId": "ServiceRegistry|temperature|1.0.0",
  "priority": 1
}
```

Listing 45: An createOrchestrationRule invocation.

3.25 PUT /api/orchestration/rules/*{id}*

Operation: changeOrchestrationRulePriority
Input: RulePriorityChange
Output: View

Changes the priority of a SimpleStore rule with `modify-priorities`, or by create and remove when `lcm.orchestration-priority-mode=replace` (not registered).

Parameters: *id*.

A successful call returns `200 OK`.

```
PUT /api/orchestration/rules/123e4567-e89b-12d3-a456-426614174000 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "priority": 2,
  "expected": {"consumer": "Consumer",
               "serviceInstanceId": "ServiceRegistry|temperature|1.0.0",
               "priority": 1}
}
```

Listing 46: An changeOrchestrationRulePriority invocation.

3.26 DELETE /api/orchestration/rules/*{id}*

Operation: removeOrchestrationRule
Output: Removal

Removes a SimpleStore rule (UUID; 400 for another identifier).
Parameters: *id*.
A successful call returns `200 OK` with the output in the body.

```
DELETE /api/orchestration/rules/123e4567-e89b-12d3-a456-426614174000 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
```

Listing 47: An removeOrchestrationRule invocation.

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "removed": true,
  "at": "..."
}
```

Listing 48: An removeOrchestrationRule response.

3.27 GET /api/logs?service=*{service}*&tail=*{tail}*

Operation: logs
Output: LogLines

The last `tail` (1 to 500, default 150) log lines of a container of the managed project, through the Docker proxy (404 for an unknown service).
Parameters: *service*, *tail*.
A successful call returns `200 OK` with the output in the body.

```
GET /api/logs?service=sysmonitor&tail=150 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 49: An logs invocation.

```
{
  "service": "sysmonitor",
  "lines": ["2026-11-09T10:00:00.000000000Z ..."],
  "collectedAt": "..."
}
```

Listing 50: An logs response.

3.28 POST /api/containers/*{service}*/*{action}*

Operation: lifecycle
Output: LifecycleResult

Starts, stops or restarts (`start`, `stop`, `restart`) a container of a service in `lcm.managed-services` through the Docker proxy: 403 outside the allowlist, 404 unknown service.

Parameters: *service*, *action*.

A successful call returns `202 Accepted` with the output in the body.

```
POST /api/containers/sysmonitor/restart HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
```

Listing 51: An lifecycle invocation.

```
{
  "service": "sysmonitor",
  "action": "restart",
  "accepted": true,
  "at": "..."
}
```

Listing 52: An lifecycle response.

3.29 GET /api/sos

Operation: sosConfigurations
Output: View

SoS compositions (all plant descriptions, up to 2000), shaped like plantDescriptions; `/api/sos/{id}` returns one composition (404 `"SoS not found"`); another path is 404 `"Unknown SoS operation"`. Not registered.

A successful call returns `200 OK`.

```
GET /api/sos HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 53: An sosConfigurations invocation.

3.30 POST /api/sos

Operation: createSoS
Input: SoSChange
Output: PlantDescription

Stores a validated SoS composition (400 with the validation messages; 428 for an active composition without `activationConfirmed`).

A successful call returns `201 Created`.

```
POST /api/sos HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

```

```
{
  "configuration": {"plantDescription": "Line 1", "active": false,
                    "include": [], "systems": [], "connections": []}
}
```

Listing 54: An createSoS invocation.

3.31 PUT /api/sos/*{id}*

Operation: replaceSoS
Input: SoSChange
Output: PlantDescription

Replaces an SoS composition when `expected` matches the stored one (409 otherwise; 400 when it includes itself).

Parameters: *id*.

A successful call returns `200 OK`.

```
PUT /api/sos/1 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "configuration": { ... },
  "expected": {"id": 1, "updatedAt": "...", ... }
}
```

Listing 55: An replaceSoS invocation.

3.32 DELETE /api/sos/*{id}*

Operation: removeSoS
Input: SoSChange
Output: Removal

Removes an SoS composition when `expected` matches the stored one.

Parameters: *id*.

A successful call returns `200 OK` with the output in the body.

```
DELETE /api/sos/1 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "expected": { ... }
}
```

Listing 56: An removeSoS invocation.

```
{
  "removed": 1
}
```

Listing 57: An removeSoS response.

3.33 GET /api/localcloud/*{route}*

Operation: localCloudConfiguration
Input: Object
Output: View

Local-cloud configuration (404 "`Local cloud configurator is not enabled`" when switched off). The route selects one operation of `localCloudConfiguration` of LocalCloudConfigurator [6], consumed through orchestration; the request body is forwarded and status and body are returned unchanged. Other routes are 404 "`Unknown local cloud configuration operation`".

Parameters: *route*.

A successful call returns `the configurator's status` with the output in the body.

```
POST /api/localcloud/plan HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "systems": ["timemanager"],
  "remove": [],
  "reapply": false,
  "smoke": true
}
```

Listing 58: An localCloudConfiguration invocation.

```
{
  "plan": {"cloud": "TestCloud", "catalogHash": "...", "systems": [ ... ],
           "steps": [ ... ], "notes": [ ... ]},
  "hash": "..."
}
```

Listing 59: An localCloudConfiguration response.

3.34 GET /api/chiplet/floorplans

Operation: chipletFloorplans
Input: Object
Output: View

Chiplet floorplan view when `lcm.chiplet-floorplans.enabled=true` (404 otherwise), through orchestrated ChipletFloorplanGenerator services: `GET` list (`page`, `size` up to 200), `/library` and `/{id}`; `POST` create (stored inactive, 201) and `POST /check-network`; `PUT /{id}` with `expectedUpdatedAt` (409 when changed); `DELETE /{id}`. Bodies carry `floorplan` with 1–64 chiplets and at most 1024 links. Not registered.

A successful call returns `200 OK`.

```
GET /api/chiplet/floorplans?page=0&size=200 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 60: An chipletFloorplans invocation.

4 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 4.17, which are used to represent things like hashes, identifiers and texts.

4.1 struct LifecycleResult

Object Field	Value Type	Description
"service"	String	Compose service.
"action"	String	`start`, `stop` or `restart`.
"accepted"	Boolean	Forwarded to the Docker proxy.
"at"	String	Time of the action.

4.2 struct LogLines

Object Field	Value Type	Description
"service"	String	Compose service.
"lines"	Array<String>	Log lines (`tail` latest).
"collectedAt"	DateTime	Collection time.

4.3 struct Health

Field	Type	Description
status	String	`ok`.
system	Name	`LocalCloudManager`.
registration	Object	`registered`, `instanceId`, `instanceIds`, `error`, `checkedAt`.

4.4 struct Login

Object Field	Value Type	Description
"password"	String	Console password (JSON only; 204 on success, 401 wrong, 429 locked).

4.5 struct OrchestrationRule

Object Field	Value Type	Description
"consumer"	String	Consumer system.
"serviceInstanceId"	String	Service instance.
"priority"	Number	Priority (1 highest).

4.6 struct PlantDescription

Object Field	Value Type	Description
"*"	Object	Plant description as in PlantDescriptionEngine `plantDescriptionManagement`.

4.7 struct Session

Object Field	Value Type	Description
"authenticated"	Boolean	Whether the request carries a valid console session.
"configured"	Boolean	Whether a console password is configured.

4.8 struct SysMonitorSubscriptionRequest

Object Field	Value Type	Description
"observationTypes"	Array<String>	Types to receive, `*` for all.
"enabled"	Boolean	Default true.

4.9 struct View

Object Field	Value Type	Description
"*"	Object	View document assembled from the providing systems; a failing source is reported in the document instead of failing the request.

4.10 struct Features

Object Field	Value Type	Description
"localCloudConfiguration"	Boolean	Local-cloud configuration view enabled.
"chipletFloorplans"	Boolean	Chiplet floorplan view enabled.
"managerName"	String	LocalCloudManager, or ChipletLocalCloudManager with floorplans.

4.11 struct ManagementPolicy

Object Field	Value Type	Description
"instanceId"	String	Policy instance (assigned by Consumer Authorization).
"provider"	String	Registered local provider.
"targetType"	String	`SERVICE_DEF` or `EVENT_TYPE`.
"target"	String	Service definition or event type.
"description"	String	At most 1000 characters.
"defaultPolicy"	Object	`policyType` (`ALL`, `WHITELIST`, `BLACKLIST`) and `policyList` of local system names.
"scopedPolicies"	Object	Per operation, a policy like `defaultPolicy` (at most 100).

4.12 struct PolicyChange

Object Field	Value Type	Description
"policy"	ManagementPolicy	Policy to grant (POST).
"expected"	ManagementPolicy	Stored policy as last read; required to replace or revoke.

4.13 struct PolicyView

Object Field	Value Type	Description
"entries"	Array	Editable local management policies.
"readOnlyEntries"	Array	Provider-owned local policies.
"collectedAt"	DateTime	Collection time.

4.14 struct Removal

Object Field	Value Type	Description
"removed"	Object	Identifier of, or `true` for, the removed entry.
"id"	String	Removed policy (policies only).

4.15 struct RulePriorityChange

Object Field	Value Type	Description
"priority"	Number	New priority, 1 to 2147483647.
"expected"	OrchestrationRule	Rule as last read (409 if changed, 428 if missing).

4.16 struct SoSChange

Object Field	Value Type	Description
"configuration"	Object	SoS model: `plantDescription` (name), `active`, `include`, `systems` with ports, `connections`.
"expected"	Object	Stored composition as last read (replace and remove).
"activationConfirmed"	Boolean	Must be `true` to save an active composition (428 otherwise).

4.17 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
Boolean	JSON Boolean, `true` or `false`.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
Name	JSON String, an Arrowhead system name.
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-LocalCloudManager-go`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. The native manager has its own browser/session, provider-adapter and optional telemetry profile. Its detailed contracts are in `Eclipse-Arrowhead-go/ah5-LocalCloudManager-go/documentation/v5.3.0`. Secure and legacy profiles must be distinguished; a console cookie is not an Arrowhead identity token.

Listening port. Default 8480 on all interfaces. Override with SERVER_PORT=N (or PORT=N) or –server.port N. Native design/route evidence: `Eclipse-Arrowhead-go/ah5-LocalCloudManager-go/documentation/LocalCloudManager`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

Primary system folder: `Eclipse-Arrowhead-rust/ah5-LocalCloudManager-rust`.

The manager has its own Cargo crate and legacy adapter profile; it is not the shared support runtime's secure binding. Its listener/provider authentication and unsupported mutations are documented in `Eclipse-Arrowhead-rust/ah5-LocalCloudManager-rust/documentation/v5.3.0`. Build from the primary Rust workspace with the applicable package/binary selection. Documentation links can resolve into the shared core/support catalogue. Independent process identity and state remain required.

Listening port. System API default 127.0.0.1:8480. Override with "listen" in CONFIG.json, SERVER_PORT=N or –server.port N. The HTTPS console listener ("consoleListen") is configured separately. Native design/route evidence: `Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/LocalCloudManager`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

6 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “localCloudManagement – Service Description (LocalCloudManager),” Eclipse Arrowhead, SD 5.3.0, 2026, localCloudManager-v5.3.0-localCloudManagement-SD.pdf.

[6] ——, “localCloudConfiguration HTTP/TLS/JSON – Interface Design Description (LocalCloudConfigurator),” Eclipse Arrowhead, IDD 5.3.0, 2026, localCloudConfigurator-v5.3.0-localCloudConfiguration-HTTP-IDD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): console operations outside the registered list (features, authorization policies, rule priority, SoS, local-cloud configuration, chiplet floorplans) added; success responses of logout, unsubscribeSysMonitor, removePlantDescription and removeOrchestrationRule corrected; error codes completed.	Claude Opus 5.5 (draft)
3	2026-11-09	5.2.2	Java-Spring implementation: Arrowhead connector 8491 (`generic_https`, `CERT_AUTH`, identity token, grant check) and console connector 8490 (session cookie, Host/Origin, proxy secret); every route with method, path, status codes and JSON from the controller; session operations console-only; `/health`; local-cloud configuration through the orchestrated `localCloudConfiguration` service. Node.js/nginx descriptions removed.	Claude Opus 5.5 (draft)
4	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	